Fiscal Year:	FY 2017	**Task Last Updated:**	FY 04/14/2017
PI Name:	DeChurch, Leslie Ph.D.		
Project Title:	Team Task Switching in Astronaut Crews on the International Space Station: Integrating Multiteam Membership, Multiteam Systems, Multitasking,& Multidimensional Networks to Monitor & Enable Functional Work Shifts in Astronaut Crews		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture (2) **Team**:Risk of Performance and Behavioral Health Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Evanston	**State:**	IL
Zip Code:	60208	**Congressional District:**	9
Comments:	NOTE: Previously at Georgia Institute of Technology until July 2016.		
Project Type:	Flight	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-ILSRA. International Life Sciences Research Announcement
Start Date:	05/12/2015	**End Date:**	10/05/2016
No. of Post Docs:	1	**No. of PhD Degrees:**	3
No. of PhD Candidates:	4	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Williams, Thomas	**Contact Phone:**	281-483-8773
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 10/5/2016 (original due date was 5/11/2018) due to PI move to Northwestern University and new award granted (Ed., 2/12/18) NOTE: Element change to Human Factors & Behavioral Performance; previously Behavioral Health & Performance (Ed., 1/17/17)		
Key Personnel Changes/Previous PI:	April 2017 report: There were no changes to key personnel named in the proposal. The following changes were made in personnel: Ivan Hernandez (post doc) was replaced with Patrick Park ; Joe McDonald (PhD candidate) was replaced with Ashley Niler.		
COI Name (Institution):	Contractor, Noshir Ph.D. (Northwestern University)		
Grant/Contract No.:	NNX15AK73G		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	We are at the dawn of a new era of human space exploration. Moving beyond low Earth orbit and the relative safety of the International Space Station (ISS) toward near-Earth asteroids and Mars presents previously unimaginable opportunities as well as organizational challenges. One significant challenge is the complexity of the operating environment within which astronauts will work. This complexity will place enormous demands on astronauts, and research is needed that develops concrete countermeasures to mitigate the risks stemming from performance decrements due to inadequate cooperation, coordination, communication, and psychosocial adaptation within a team. Astronauts will push the bounds of human cognitive and social functioning as they multitask across tasks, teams, and tools working toward personal, team, and system goals. This multidisciplinary research project is designed to help them do just that. This three-year programmatic investigation into team task switching leverages: (1) agent-based models to understand how task switching behavior and performance-related switching costs occur based on dynamic interplay between independent and interdependent tasks, (2) laboratory experiments conducted in two multiteam systems laboratories (one at Georgia Tech, the other at Northwestern) to test the theoretical model derived from agent-based models, (3) unobtrusive data collection strategies to capture information on the tasks and complex social network structures of ISS crew members, (4) design of interventions to enable adaptive team task switches using virtual experiments, (5) validation of intervention strategies using one of NASA's space analogs, and (6) development of a dashboard decision aid to anticipate and pre-empt dysfunctional task switching. An innovative feature of our investigation is the use of a multidimensional network approach to characterize and model the switches between tasks, tools, teams, and multi-team systems. The combined outputs of this proposed multi-disciplinary project speak directly to NASA Human Research Program's (HRP's) identified team gaps for autonomous, long duration, and/or long distance exploration missions associated with the need to identify (1) key threats to the team (Team Gap 1), (2) countermeasures to support team function (Team Gap 3), (3) validated methods that can be used to maintain team function (Team Gap 5), and (4) psychological and psychosocial factors, measures, and combinations thereof that can be used to compose highly effective crews (Team Gap 8). NOTE: Project continues as "Team Task Switching in Astronaut Crews on the International Space Station: Integrating Multiteam Membership, Multiteam Systems, Multitasking, & Multidimensional Networks to Monitor & Enable Functional Work Shifts in Astronaut Crews--80NSSC18K0276," due to Principal Investigator move to Northwestern University in Fall 2016.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Previous models of task switching revolve around task and individual factors that affect work efficiency as individuals move from task to task throughout their workday. However, individuals not only accomplish tasks individually using one technological tool, they do so as members of multiple teams using a variety of technologies. Our research on team task switching examines factors that affect the decision and ability of individuals to switch between tasks with varying levels of interdependence, as well as discuss the mediators and moderators that determine adaptive switching in today's complicated work environments. This model affords a comprehensive understanding of how individuals adapt to dynamic, environmentally-triggered performance demands requiring them to change tasks, teams, and technologies, and shift back and forth between personal, team, and system goals. Our research will be useful for investigating task switching in any modern-day organization facing complex collaborative challenges, such as NASA space exploration, large scientific consortia (e.g., CERN-the European Organization for Nuclear Research), cybersecurity teams, healthcare systems, and the military. Furthermore, findings could then be leveraged to develop system-wide interventions that increase overall work efficiency and resilience in safety-critical systems.
	Over the course of this three-year programmatic research program, we (1) explore issues that can arise when astronauts switch between tasks, teams, and technologies within the complex, multiteam systems that characterize operations on the International Space Station (ISS) and during Long Duration Space Exploration (LDSE), as well as (2) investigate the effects of astronaut task, team, and technology switches on individual-, team-, and system-level performance. In particular, astronauts work individually as well as within and across different teams to accomplish their work. They also complete a variety of tasks and engage a variety of technologies in the process. Shifting across these tasks, teams, and tools holds the potential for "switch costs" in the form of process loss and performance decrements. Aspects of the various tasks, teams, and technologies make switching to/from each more/less seamless. We explore the implications of effective/efficient switching. In Year 1 we developed a conceptual framework to guide our thinking in this area, and have conducted research in Year 2, which allows us to begin to draw conclusions about the role of task/team/technology switches on individual/team/multiteam process and performance. Our conceptual framework highlights three key types of switches (task, team, and technology) as well as two directions of switching (lateral versus vertical). Lateral switches occur when individuals shift across tasks, teams, or technologies/tools. Such lateral shifts can be either singular or compound. A singular shift entails changes in one of the three elements (e.g., the individual switches from working on task A to task B or from working with tool A to tool B). In contrast, a compound shift entails changes on two or more of the task, team, and tool dimensions (e.g., a compound task-team shift means the individual goes from working on task A with individuals X, Y, and Z to working on task B with individuals I, J, and K). Our framework also describes vertical switches, which involve a shift in the level of interdependence (e.g., individual, single team, multiple teams) needed to complete a task. Vertical shifts can be either upward (e.g., working independently to working with a team) or downward (e.g., working with a team to working independently). In Year 2 of our project, we've conducted a five-pronged effort at understanding the potential role of team/task/tool switching in long duration space exploration: (1) policy-capturing research to explore the factors that affect attraction to different teams, (2) laboratory studies to explore ability to effectively make switches, (3) agent based model development to explore the psychological, task, tool, and social network based factors that determine desire to stay with or switch from a task/team/tool, (4) analog studies using HERA (Human Exploration Research Analog) crews to explore how task characteristics affected vertical and lateral switches, and (5) a task switching survey aboard the ISS. The policy capturing research conducted in Study 1 explores the team characteristics that may contribute to team

Task Progress:	attractiveness. We explored the extent to which aspects of team cognition, cohesion, and coordination made teams more or less attractive. Results suggested team cohesiveness was the most influential factor in determining the extent to which a team was deemed “attractive” to work with. Team attractiveness is one factor that will affect an astronaut’s desire to make a team switch. In Study 2, we conducted a series of laboratory studies to explore team and tool characteristics (e.g., shared team cognition and tool availability) that affect switch preferences and efficiency. We employed laboratory-based survival simulation tasks to explore stickiness and attractiveness of teams/tasks/tools as well as the ability to easily switch across tasks/teams. Results suggested (1) shared mental models within the team (shared cognition) predict efficiency of switching across teams/tasks, and (2) team cohesion predicts the extent to which an existing versus alternative team is more attractive to work with. Understanding factors that affect switch efficiency and attractiveness of alternative teams/tasks will inform future models aimed at developing work schedules in similarly complex environments. In Study 3, we have worked to develop an agent based model (ABM) to predict the extent to which factors like psychological, task, tool, and/or social network attributes predict preference to stay with an existing task versus switch to an alternative task. We developed the ABM in two phases. First, we canvassed the relevant literature on task switching to determine which factors may be important to determining the “stickiness” of a task. Second, we collected laboratory data to estimate and then test the parameters of the agent based model. The ABM will be used in Year 3 to run “virtual experiments” wherein we can pose “what if” questions that will directly inform the development of effective task switching schedules. In Study 4, we collected analog data in NASA’s HERA analog to explore the factors (like task difficulty, interest, salience, and interdependence) that influence preference and efficiency in vertical and horizontal switches. We explored how personality, cognitive similarity, and team dynamics and structure predict the relative stickiness and attractiveness among individuals within a team and multiteam system. In Study 5, we designed a survey to administer aboard the ISS. Thus far we have designed and pilot tested the survey, and then completed the various stages of vetting for flight selection. As of the end of this reporting period we have just learned that our study was selected for flight! This effort will be conducted next year. Our research will provide fundamental, generalizable findings in the area of efficient task switching across teams and multiteam systems. Specifically, previous models of task switching revolve around task and individual factors that affect work efficiency as individuals move from task to task throughout their workday. However, individuals not only accomplish tasks individually using one technological tool, they do so as members of multiple teams using a variety of technologies. Our conceptual framework of team task switching lays out a framework of vertical and lateral shifts, the sources of inertia, and the mediators and moderators that determine adaptive switching in today’s complicated work environments. This model affords a comprehensive understanding of how individuals adapt to dynamic, environmentally-triggered performance demands requiring them to change tasks, teams, and technologies, and shift back and forth between personal, team, and system goals. Our framework will be useful for investigating task switching in any modern-day organization facing complex collaborative challenges, such as NASA space exploration, large scientific consortia (e.g., CERN), cybersecurity teams, healthcare systems, and the military. Furthermore, findings could then be leveraged to develop system-wide interventions that increase overall work efficiency and resilience in safety-critical systems. In this review period, we have collected data across multiple platforms to begin to understand the effects of variables across our conceptual framework on individual and team task switching performance, preferences, and motivations. In addition to providing Earth Benefits in the area of efficient task switching across teams, tasks, and technologies, we are also testing our hypotheses in analogs that recreate the unique contexts in which astronauts operate (i.e.,extreme, isolated/confined environments). We will compare results across research paradigms to see how effects of isolation/confinement may moderate team task switching behavior and performance, thus providing unique findings that will inform NASA’s operations aboard the International Space Station (ISS) and on Long-Duration Space Exploration (LDSE) missions. NOTE: Project continues as "Team Task Switching in Astronaut Crews on the International Space Station: Integrating Multiteam Membership, Multiteam Systems, Multitasking, & Multidimensional Networks to Monitor & Enable Functional Work Shifts in Astronaut Crews--80NSSC18K0276," due to Principal Investigator move to Northwestern University in Fall 2016. See that project for subsequent reporting.
Bibliography Type:	Description: (Last Updated: 11/17/2025)
Abstracts for Journals and Proceedings	McDonald JD, DeChurch LA. "What makes teamwork attractive? A policy capturing study to identify influential social factors in team task switching." In McDonald, J. D., Jones, B. R., and Contractor, N. S. (co-chairs), The Many “I”s of a Team: Person Centric Perspectives on Teams. Symposium at 31st Annual Conference of the Society for Industrial and Organizational Psychology, Anaheim, CA, April 14-16, 2016. 31st Annual Conference of the Society for Industrial and Organizational Psychology, Anaheim, CA, April 14-16, 2016. , Apr-2016
Abstracts for Journals and Proceedings	DeChurch LA, Plummer G, Contractor NS, Mesmer-Magnus JR. "The costs of switching between team and multiteam tasks and the role of shared cognition." Poster presented at 2017 NASA Human Research Program Investigators’ Workshop, Galveston, TX, January 23-26, 2017. Abstracts. 2017 NASA Human Research Program Investigators’ Workshop, Galveston, TX, January 23-26, 2017. , Jan-2017
Abstracts for Journals and Proceedings	Park P, DeChurch LA, Contractor NS. "Understanding elective task switching." Poster presented at 2017 NASA Human Research Program Investigators’ Workshop, Galveston, TX, January 23-26, 2017. Abstracts. 2017 NASA Human Research Program Investigators’ Workshop, Galveston, TX, January 23-26, 2017. , Jan-2017
Abstracts for Journals and Proceedings	DeChurch L, Larson L, Jones B, Contractor N, Johnson J. "Leadership emergence in space multiteam systems." 2017 NASA Human Research Program Investigators' Workshop, Galveston, TX. January 23-26, 2017. 2017 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 23-26, 2017. , Jan-2017

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Dissertations and Theses	Plummer G. "The costs of switching between team and multiteam tasks, and the role of shared cognition." Unpublished Master's thesis, School of Psychology, Georgia Institute of Technology, Atlanta, GA, November 2016. , Nov-2016
Papers from Meeting Proceedings	McDonald JD, LoPilato AC, Thomas RP, DeChurch LA, Contractor NS. "What makes teamwork attractive? A policy capturing study." Paper presented at the 31st Annual Conference of the Society for Industrial and Organizational Psychology, Anaheim, CA, April 14-16, 2016. 31st Annual Conference of the Society for Industrial and Organizational Psychology, Anaheim, CA, April 14-16, 2016. , Apr-2016
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